

1 Solar System Model

Planetary dynamics and Fourier analysis

1.1 Keplerian orbits, perihelion and aphelion

Exercise 1. What can be inferred from `orbits_keplerian.png`, and what is difficult to judge from a single three-dimensional view?

Model answer. The trajectories are closed, approximately elliptical orbits around the Sun, with different orbital sizes and different orientations in three-dimensional space. The outer planets dominate the plot scale, so the inner orbits are strongly compressed near the centre. A single perspective view is therefore poor for judging the eccentricity of the inner planets, their exact semi-major axes, small inclinations, and precise relative orientations. Orbital periods cannot be read reliably from this static trajectory plot.

Exercise 2. Estimate Pluto's orbital period from the repeating Sun–Pluto distance signal.

Model answer. The repeating interval is about 2.5×10^2 yr, approximately 248 yr. This agrees with the value in the practical table.

Exercise 3. How is Earth's lower eccentricity reflected in the Sun–planet distance variation?

Model answer. Earth's Sun–planet distance varies by only a small fraction of its mean value, so the oscillation has a small amplitude. Pluto has a much more eccentric orbit and consequently shows a much larger difference between perihelion and aphelion.

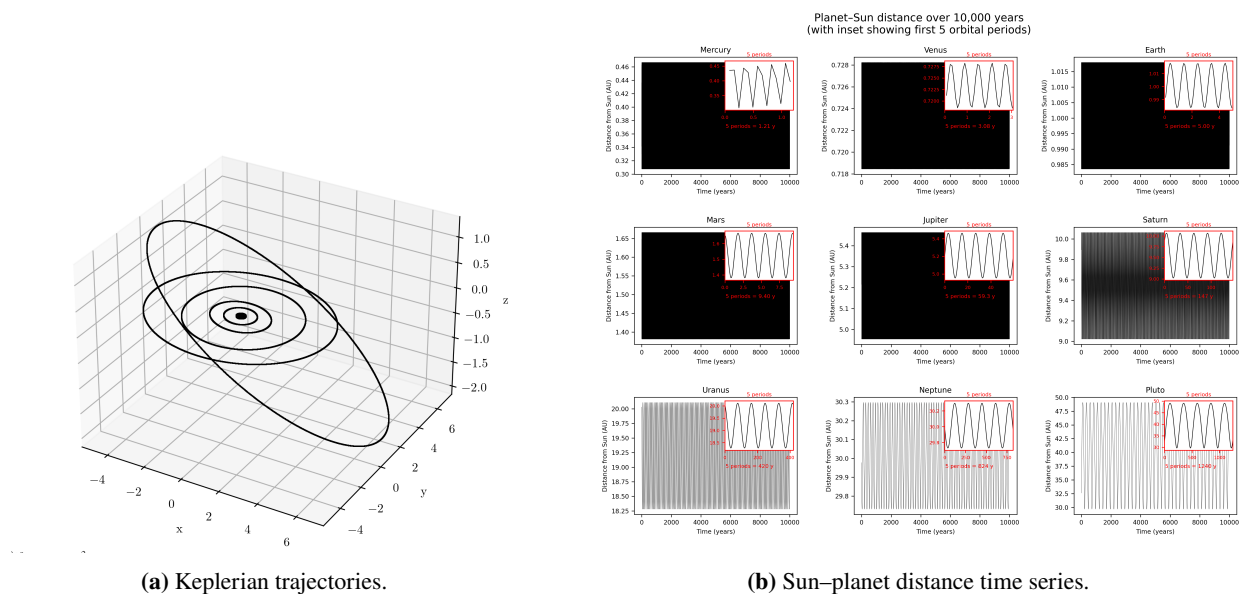


Figure 1: The trajectory and distance diagnostics emphasise different aspects of the same orbital motion.

1.2 Fourier analysis: finding periodic behaviour

Exercise 4. Identify the dominant Fourier periods and explain why the outer-planet peaks are harder to estimate precisely.

Model answer. The strongest peaks correspond closely to the orbital periods: Mercury ≈ 0.24 yr, Venus ≈ 0.62 yr, Earth ≈ 1.0 yr, Mars ≈ 1.9 yr, Jupiter ≈ 11.9 yr, Saturn ≈ 29.5 yr, Uranus ≈ 84 yr, Neptune ≈ 165 yr and Pluto ≈ 248 yr. The frequency resolution of a finite record is roughly fixed, $\Delta f \sim 1/T_{\text{record}}$, but converting frequency to period gives an uncertainty that grows approximately as $T^2 \Delta f$. The long-period outer planets also complete fewer cycles within the integration. Their peak periods are therefore less sharply constrained.

Exercise 5. In the Earth zoom, identify smaller peaks and test whether their frequencies are close to integer multiples of the fundamental.

Model answer. The principal Earth peak is near $T = 1$ yr. Smaller peaks occur at about 0.5, 0.33 and 0.25 yr. Their frequencies are approximately 1, 2, 3 and 4 yr^{-1} . Dividing by the lowest frequency therefore gives ratios close to 1 : 2 : 3 : 4. These are harmonics of the orbital-period signal.

Exercise 6. Why can the planet–Sun distance contain harmonics as well as the fundamental orbital frequency?

Model answer. A Fourier series contains only one frequency for a perfect sinusoid. The radial distance around an eccentric Keplerian orbit is periodic but is not an exact sine wave when expressed as a function of time. A non-sinusoidal periodic waveform is represented by a fundamental plus harmonics at integer multiples of that frequency.

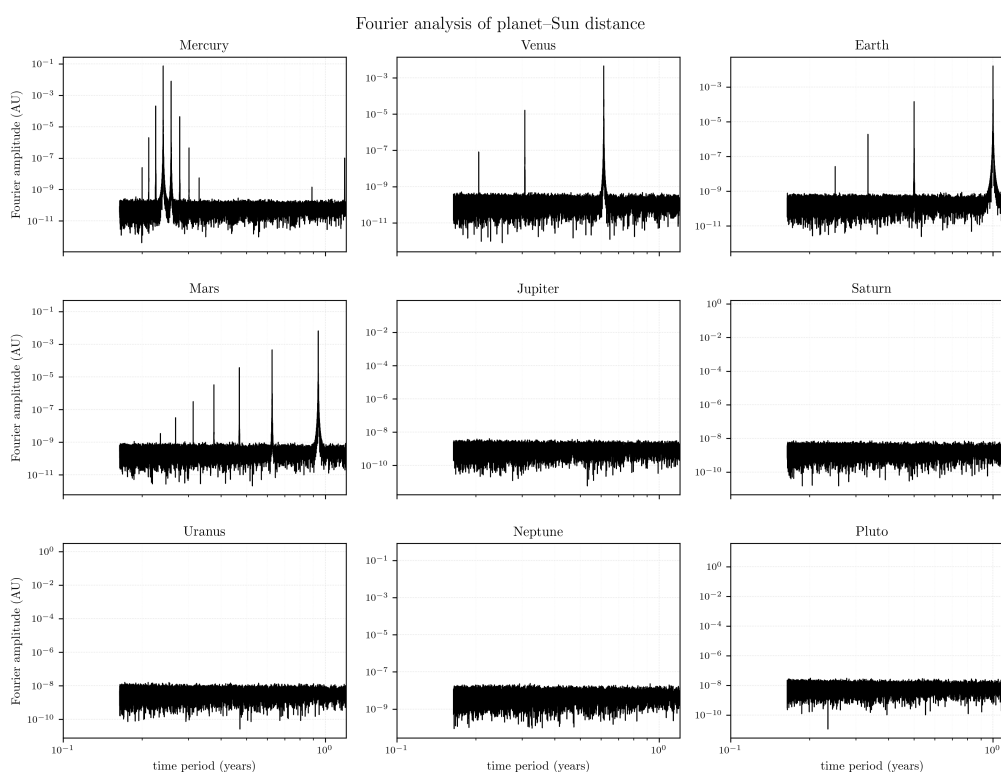


Figure 2: The inner-planet spectra on the 0.1–1.2 yr scale. Earth’s main 1 yr peak is accompanied by harmonics near 1/2, 1/3 and 1/4 yr.

1.3 Mutual planetary interactions

Exercise 7. Compare the Saturn spectrum in the Sun-only and interacting calculations.

Model answer. The Sun-only spectrum is dominated by Saturn’s own orbital periodicity and its harmonics; on the 10–30 yr scale this includes the strong feature near 29.5 yr and a harmonic near 14.7 yr. When mutual interactions are enabled, additional peaks and much richer spectral structure appear. In particular, a strong feature appears near 20 yr. This feature is absent or much weaker in the independent two-body calculation and is associated with the Jupiter–Saturn interaction timescale.

Exercise 8. Calculate the Jupiter–Saturn synodic period and explain why an interaction generates this new timescale.

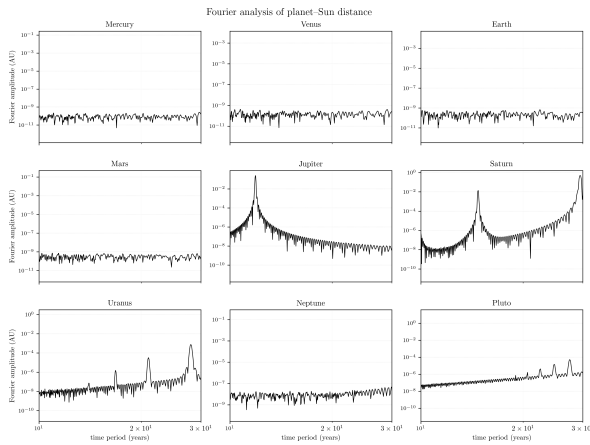
Model answer. Using

$$\frac{1}{T_{\text{syn}}} = \left| \frac{1}{11.9} - \frac{1}{29.5} \right|,$$

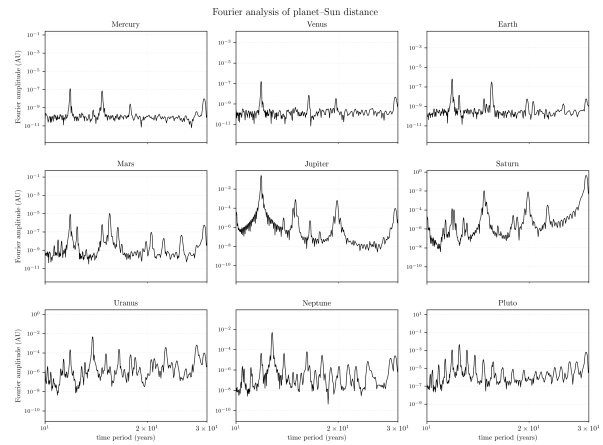
gives

$$T_{\text{syn}} \approx 19.95 \text{ yr},$$

which is consistent with the feature near 20 yr. Two bodies with different orbital frequencies repeatedly move into and out of the same relative configuration. Their relative phase therefore varies at the difference frequency $|f_J - f_S|$, producing the synodic or beat period. Mutual gravitational perturbations can imprint this combination frequency on the orbital signals.



(a) Sun-only reference.



(b) All bodies interacting.

Figure 3: Saturn spectra on identical 10–30 yr axes. Mutual interactions add combination frequencies, including the Jupiter–Saturn synodic timescale near 20 yr.

1.4 The Sun’s wobble and exoplanets

Exercise 9. Which periodicities dominate the Sun’s barycentric motion, and which planets have the largest influence?

Model answer. The strongest short-to-intermediate-period signal is close to Jupiter’s 11.9 yr orbit, with a strong contribution near Saturn’s 29.5 yr period. Longer-period features associated with Uranus and Neptune are also visible. Jupiter is the largest single contributor, with Saturn also important; the other giant planets contribute appreciably at longer periods. The terrestrial-planet signals are much smaller.

Exercise 10. Use $r_\star \sim (m_p/M_\odot)a_p$ to explain why giant planets dominate stellar reflex motion.

Model answer. The stellar displacement increases with both planet mass and orbital size. Giant planets have masses tens to hundreds of times larger than the terrestrial planets, and they also orbit much farther from the Sun. The product $m_p a_p$ is therefore much larger, giving a correspondingly larger displacement of the star around the system barycentre. This is the physical basis of astrometric stellar wobble and is closely related to radial-velocity exoplanet detection.

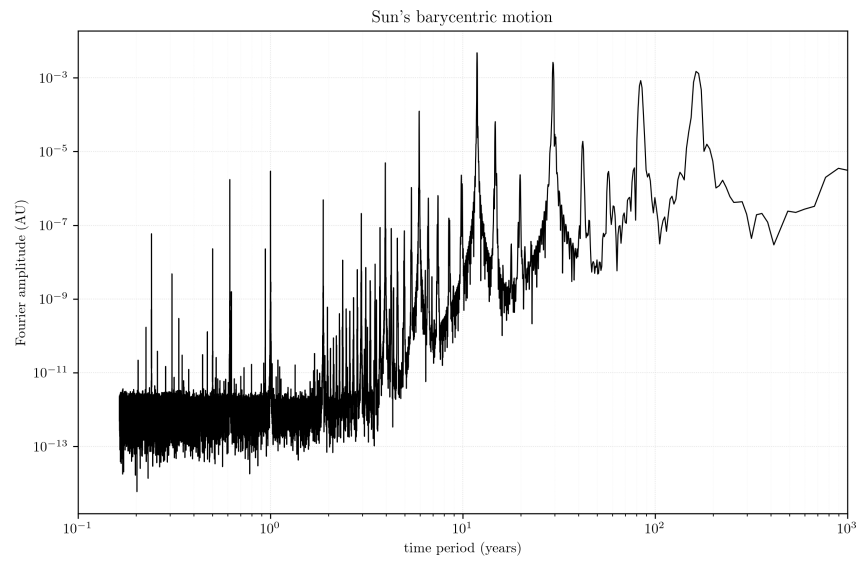


Figure 4: Fourier spectrum of the Sun's barycentric motion. The giant-planet timescales dominate.